

2018 Chemistry First Round ESSLCE Online Exam



- The masses of different substances, liberated or deposited by the same amount of electricity, are proportional to their equivalent masses. This statement is _____.
 - Faraday's first law of electrolysis
 - the law of definite proportion
 - Faraday's second law of electrolysis
 - the law of multiple proportion
- An unknown element has the following electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$. To which period and group does this element belong in the periodic table, respectively?
 - 3 and VA
 - 5 and IIIA
 - 3 and IIIA
 - 4 and IIA
- Which of the following reaction represents the chemical equation for the reaction that takes place between aluminum metal and solution of hydrochloric acid?
 - $2AlCl_3(aq) + 3H_2(g) \rightarrow 2Al(s) + 6HCl(aq)$
 - $2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$
 - $2Al + 6HCl(aq) \rightarrow 2AlCl_3(aq)$
 - $Al + HCl \rightarrow AlCl_3$
- A cell containing $AlCl_3$ as an electrolyte, is placed in series with another cell containing $AgNO_3$ as an electrolyte. What mass of Ag is deposited, when 18g of Al is deposited at the cathode electrode? ($M_{Ag}=108g/mol$ and $M_{Al}=27g/mol$)
 - 36g
 - 24g
 - 216g
 - 72g
- Nitric acid is a colorless fuming liquid with a boiling point of $83^\circ C$. However, it turns yellow as it stays for long time. Which of the following accounts for the observed property of nitric acid?
 - NO_3^-
 - NO_2

c. NO₂–

d. NO

6. Which of the following tells the exact number of atoms of different elements present in a molecule?

a. Molecular mass

b. Empirical formula

c. Formula mass

d. Molecular formula

7. What is the concentration of hydrogen ion in a solution of 0.1 M nicotinic acid, HC₆H₄NO₂, at 25°C? ($K_a = 1.4 \times 10^{-5}$)

a. 1.2×10^{-5} M

b. 1.2×10^{-3} M

c. 1.4×10^{-5} M

d. 1.4×10^{-6} M

8. Chlorine has two naturally occurring isotopes. In a sample of chlorine, 25% of the atoms are Cl-37 and 75% of the atoms are Cl-35. What is the average atomic mass of chlorine?

a. 26.25

b. 25.75

c. 35.5

d. 36.0

9. A student performs an experiment to distinguish different oxides by doing the following experiment. He/she put 100 mL KI solution in to four different test tubes. He/she acidified each of these solutions with 10 drops of dilute H₂SO₄ solution. He/she added a spatula full of CaO in the first, Al₂O₃ in the second, Na₂O₂ in the third and P₄O₁₀ in the fourth test tubes. Finally, he/she added 10 mL starch solution to each of the four test tubes. In which test tube will he/she observe a color change? In the test tube that contains

a. Test tube 2

b. Test tube 1

c. Test tube 3

d. Test tube 4

10. Combustion of acetylene at about 3000°C with oxygen produces an intense hot flame. In which group of hydrocarbons is this compound classified?

a. Alkanes

b. Aromatics

c. Alkenes

d. Alkynes

11. Phosphorus pentafluoride, PF₅ is one example of exception to the octet rule. Why does PF₅ is considered to be exceptional to the octate rule? This is because the

a. central atom has more than eight valence electrons.

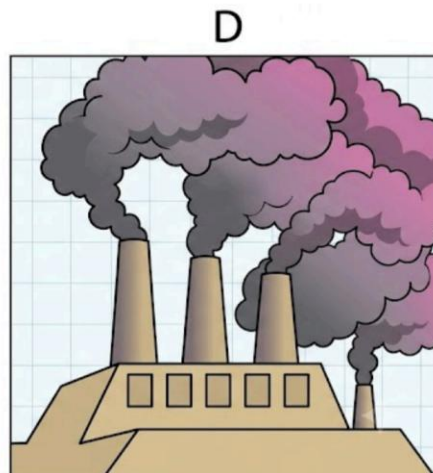
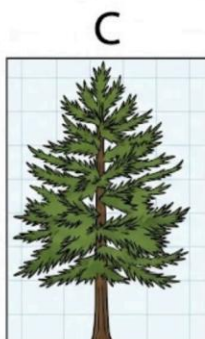
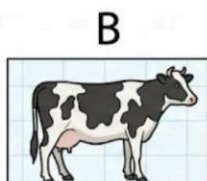
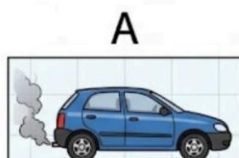
b. atoms attached to P have more than eight valence electrons.

c. central atom has less than eight valence electrons.

d. central atom has eight valence electrons.

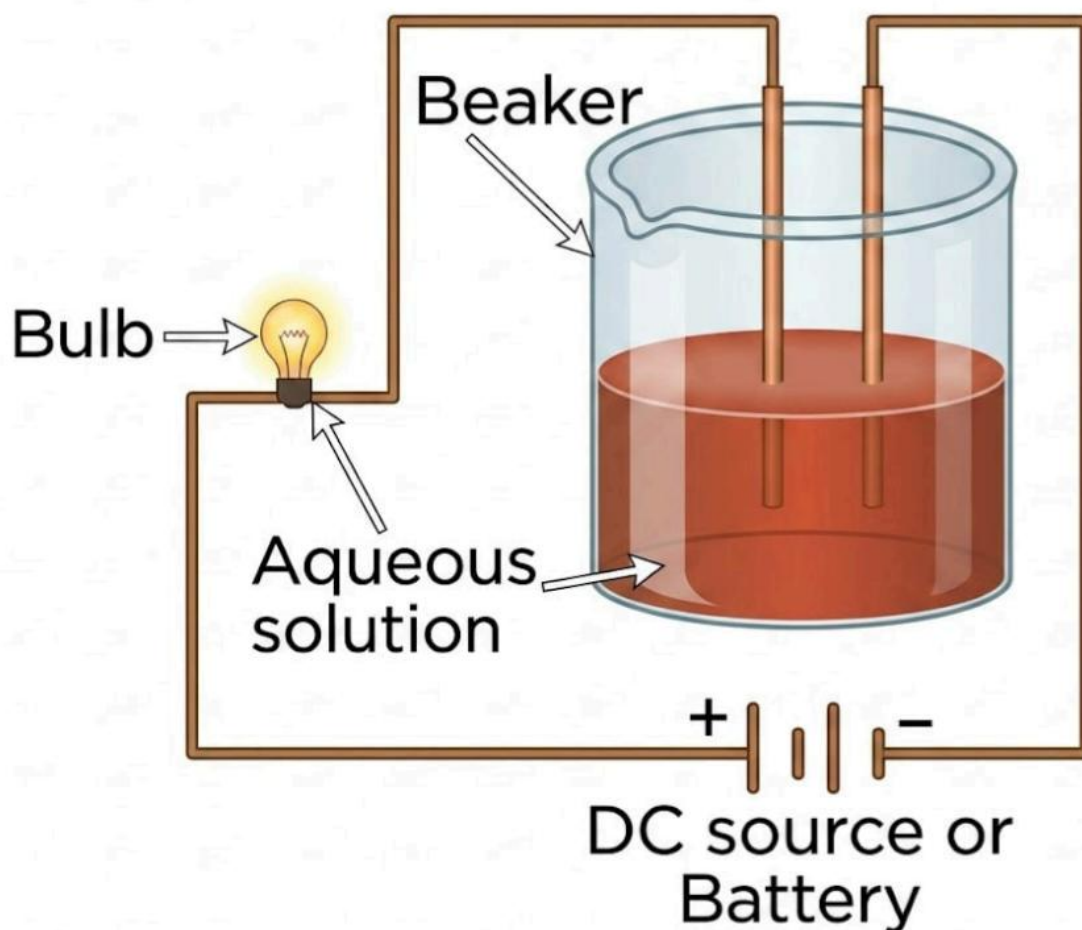
12. Which of the above does NOT contribute to global warming?

Atmospheric Carbon Dioxide



- a. B
- b. A
- c. D
- d. C

13. You are supposed to test the conductivity of the substances using a conductivity apparatus. Which of the following aqueous solutions does NOT cause the bulb to glow when the switch turns on?



- a. Sodium hydroxide solution
- b. Copper sulphate solution
- c. Table salt solution
- d. Sugar solution

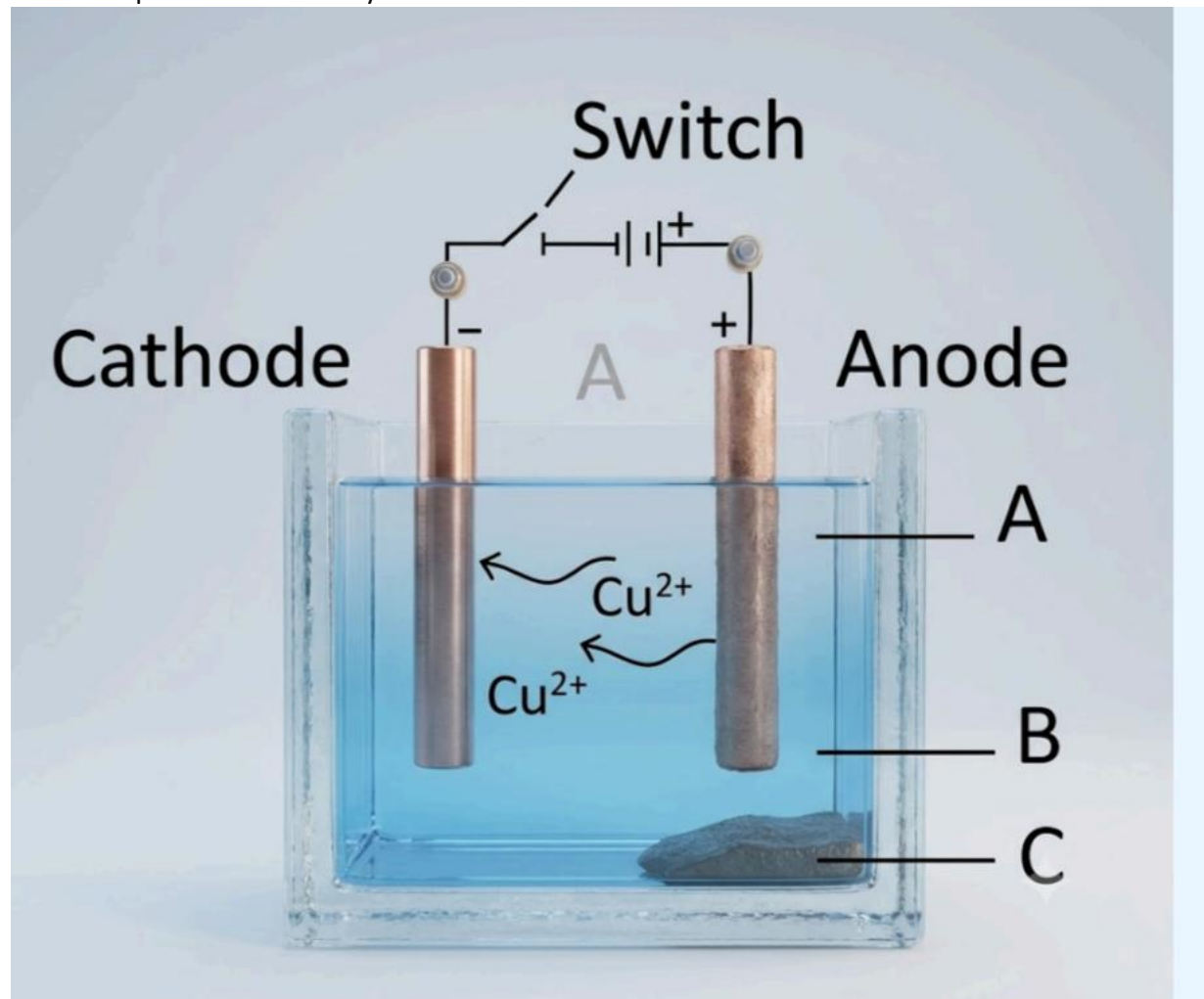
14. Copper metal reacts with sulfuric acid to produce copper sulphate, sulfur dioxide and water. Which of the following is the coefficient of H_2SO_4 after balancing this chemical reaction?

- a. 6
- b. 1
- c. 2
- d. 3

15. What is the amount of energy emitted when an electron moves from the $n=4$ to $n=2$ energy level? ($R_H=2.18 \times 10^{-18}\text{J}$)

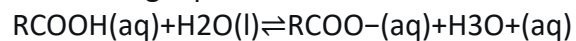
- a. $-6.54 \times 10^{-18}\text{J}$
- b. $-4.09 \times 10^{-19}\text{J}$
- c. $4.09 \times 10^{-19}\text{J}$
- d. $6.54 \times 10^{-18}\text{J}$

16 Consider the following diagram used for the electrolytic refining of copper. What is the name of species indicated by letter 'C'?



- a. Copper sulphate solution
- b. Strip of pure copper
- c. Cathode mud
- d. Anode mud

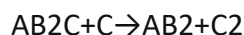
17. Consider the following equilibrium reaction of a certain weak acid, RCOOH :



Which of the following is the CORRECT expression of the percent ionization of RCOOH ?

- a. $\frac{\text{Equilibrium concentration of } \text{H}_3\text{O}^+}{\text{Initial concentration of the acid}} \times 100$
- b. $\frac{\text{Initial concentration of the acid}}{\text{Equilibrium concentration of the acid}} \times 100$
- c. $\frac{\text{Initial concentration of } \text{RCOO}^-}{\text{Initial concentration of the acid}} \times 100$
- d. $\frac{\text{Equilibrium concentration of the acid}}{\text{Initial concentration of the acid}} \times 100$

18. Consider the reaction given below:



Which of the following successful collisions are effective in the formation of the products?

- The collision that results in the A and C atoms are directly facing each other.
- The collision that results in the C atoms are directly facing each other.
- The collision that results in the B and C atoms are directly facing each other.
- The collision that results in the A and B atoms are directly facing each other.

19. At 0°C and 1.0 atm, the solubility of O₂ gas in water is 48.4 mole per liter. What is the molar solubility of O₂ in a saturated water solution at a partial pressure of 0.20 atm?

- $1.20 \times 10^{-3} \text{ M}$
- 9.68 M
- $4.10 \times 10^{-3} \text{ M}$
- 2.44 M

20. Which one of the following does NOT explain the chemical properties of acidic oxides?

Acidic oxides react with:

- acids to form salt and water.
- basic oxides to form salts.
- water to form acids.
- base to form salt and water.

21. What is the electronic configuration of an element with atomic number of 19, on the main energy levels?

- 2,8,7,2
- 2,8,9
- 2,9,8
- 2,8,8,1

22. Recently, Lake Tana was invaded by a surface water plant commonly known as "Emboch". Which one of the following describes this phenomenon?

- Industrial pollution
- Water pollution
- Soil pollution
- Air pollution

23. Which one of the following philosophers believed that matter is NOT continuous?

- Plato

- b. Democritus
- c. Empedocles
- d. Aristotle

24. Which of the following oxides has oxygen with negative one (–1) oxidation state in its molecule?

- a. Basic oxide
- b. Super oxide
- c. Acidic oxide
- d. Peroxide

25. Combustion of methane results in a release of heat. Which of the following statements describe these chemical phenomena? The release of heat is attributed to the energy change:

- a. due to the emission of light.
- b. that accompanies the reaction process.
- c. that resists changes in the reaction process.
- d. due to the absorption of light.

26. Which one of the following is a synthetic polymer?

- a. Teflon
- b. Enzymes
- c. Starch
- d. Cotton

27. Which one of the following is a polar covalent molecule?

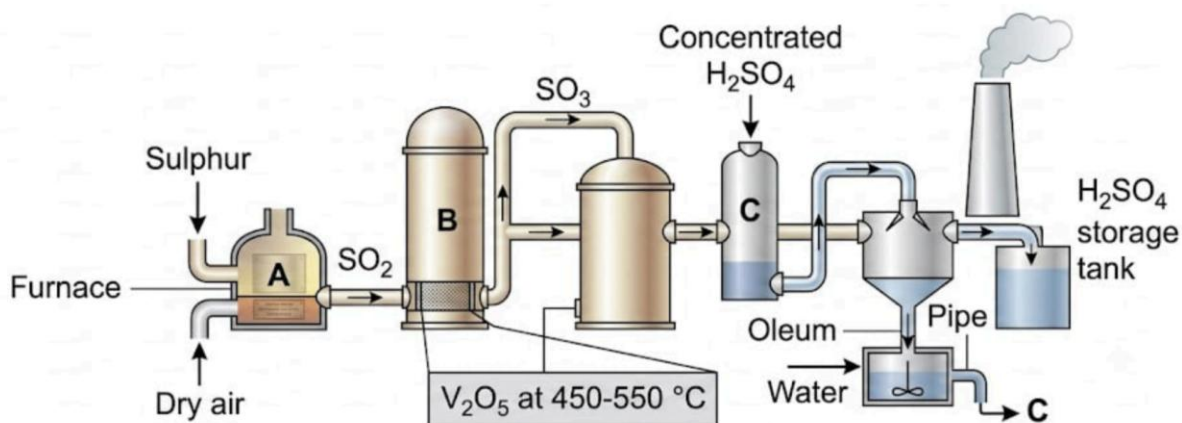
- a. CO₂
- b. CCl₄
- c. CH₄
- d. SO₂

28. Consider the following electrochemical

reaction: $\text{Cu(s)} + 2\text{Ce}^{4+}(\text{aq}) \rightleftharpoons \text{Cu}^{2+}(\text{aq}) + 2\text{Ce}^{3+}(\text{aq})$ Which of the following effects increases the cell potential of the reaction?

- a. Decreasing concentration of Cu(s)
- b. Increasing concentration of Cu^{2+}
- c. Increasing concentration of Ce^{3+}
- d. Increasing concentration of Ce^{4+}

29. In this schematic diagram, what are the compounds represented by letters A, B and C, respectively?



- SO_2 , H_2SO_4 and SO_4^{2-}
- SO_3 , SO_2 and H_2SO_4
- SO_2 , SO_3 and H_2SO_4
- SO_2 , H_2SO_4 and SO_3

30. Which one of the following is an alloy of 87% copper, 10% tin and 3% zinc?

- Solder
- Gun metal
- Bronze
- Gold alloys

31. What is the pH of a solution when the concentration of hydroxide ion is $1.0 \times 10^{-9} \text{ M}$?

- 5
- 1.8
- 9
- 7

32. A chemistry teacher designed to fill three similar balloons with three different gases to compare the rate of diffusion of the gases. He/she filled the first balloon with helium gas, the second with hydrogen gas and the third with oxygen gas. Which balloon do you think fill first?

- Second balloon
- Third balloon
- First balloon
- First and second balloon fills equal

33. Oxygen reacts with sulfur to form sulfur dioxide. Experimentally, it was formed that the total mass of the reactants (oxygen and sulfur) is equal to the mass of the product (sulfur dioxide). Which of the following laws illustrate this experiment? The law of:

- constant composition
- conservation of mass
- multiple proportions
- definite proportions

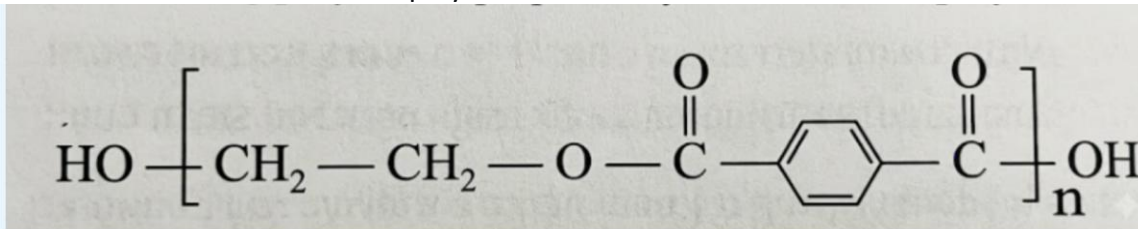
34. Which of the following is formed from the combinations of two or more substances with at least two visible phases?

- a. Heterogeneous mixture
- b. Compound
- c. Homogeneous mixture
- d. Solution

35. If a 50 cm^3 sample of gas exerts a pressure of 60.0 kPa at 35°C , what volume will it occupy at STP (0° and 1 atm) in cm^3 ? ($1 \text{ atm} = 101.325 \text{ kPa}$)

- a. 226.86
- b. 234.0
- c. 26.24
- d. 265.0

36. The following polymer is prepared by condensation polymerization. What are the monomers for this condensation polymer?



- a. Adipic acid and hexamethylene diamine
- b. Ethylene glycol and terephthalic acid
- c. Ethylene glycol and adipic acid
- d. Terephthalic acid and hexamethylene diamine

37. Which of the following is the molecular formula of 2-hexanol?

- a. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- b. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{C}(\text{OH})_2\text{CH}_3$
- c. $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}(\text{OH})_2\text{CH}_2\text{CH}_3$
- d. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

38. Crystalline solids are classified as covalent, ionic, molecular and metallic crystals. Which of the following is an example of a metallic crystal?

- a. Diamond
- b. Gold
- c. KCl
- d. CaCO_3

39. A Grade-12 student is allowed to set-up a Zn-Cu cell containing 1.0 M ZnSO_4 solution and 1.0 M CuSO_4 solution to measure a cell potential using voltmeter. What will be the reading of the voltmeter for his/her experiment?

- a. 0.34 V
- b. 0.76 V
- c. 0.42 V
- d. 1.1 V

40. Which of the following factors does NOT affect preferential discharge?

- a. Nature of ions
- b. Nature of electrodes

c. Thickness of the electrodes

d. Concentration of ions

41. A given element 'M' has atomic number of 20. What is the common ion form of this element after ionization process?

a. M^+

b. M^{2+}

c. M^-

d. M^{2-}

42. Which one of the following periods, is the short period in the modern periodic table?

a. Period 5

b. Period 3

c. Period 4

d. Period 6

43. What is the reaction type that involves the breaking down of a single compound into two or more components?

a. Decomposition reaction

b. Redox reaction

c. Displacement reaction

d. Combination reaction

44. A student carried out an experiment to test the reaction between silver nitrate solution and dilute hydrochloric acid under optimum reaction conditions. Which one of the following reactions will be obtained according to the law of conservation of mass?

a. $HCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + HNO_3(l)$

b. $HCl(aq) + AgNO_3(aq) \rightarrow 2AgCl(s) + HNO_3(l)$

c. $HCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + 2HNO_3(l)$

d. $2HCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + HNO_3(l)$

45. Which of the following is a non-renewable natural resource?

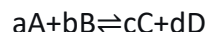
a. Cotton

b. Gold

c. Silk

d. Wood

46. Consider the following equilibrium reaction:



Which of the following is CORRECT about the given reaction?

a. Removing both reactants and products decrease the value of K_c .

b. Adding products to the system decreases the value of K_c .

c. Adding reactants to the system increases the value of K_c .

d. K_c is not affected by removing both reactants and products.

47. What is the molecular shape of phosphorus trichloride, PCl_3 ? (Atomic number P=15, Cl=17)

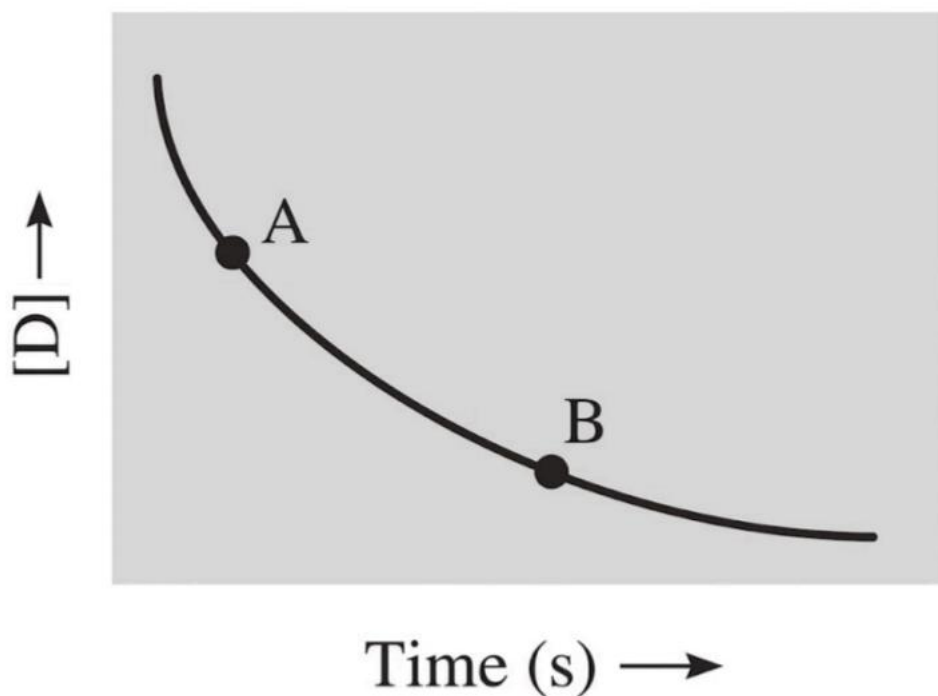
a. Trigonal planar

b. Trigonal pyramidal

c. Square planar

d. Tetrahedral

48. Which of the following is CORRECT about the rate of reaction of reactant D?

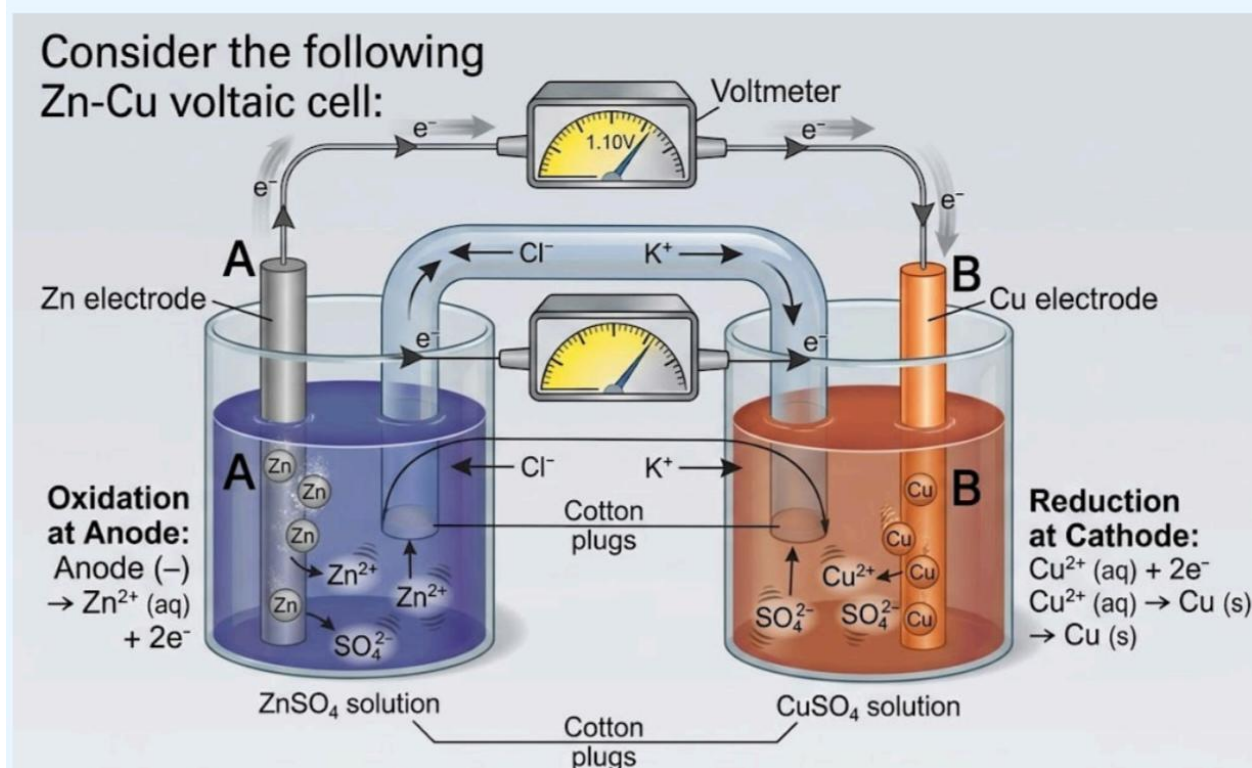


- a. The rate for this reaction is constant at all points
- b. The instantaneous rate at point A is greater than at point B
- c. The instantaneous rate at point A is equal to that at point B
- d. The instantaneous rate at point A is less than at point B

49. Neutron is one of the fundamental sub-atomic particles of an atom. What is the relative charge of a neutron in an atom?

- a. 0
- b. +1
- c. -1
- d. +2

50. In the above voltaic cell, what does letter B represent?



- Cathode
- Voltmeter
- Anode
- Salt bridge

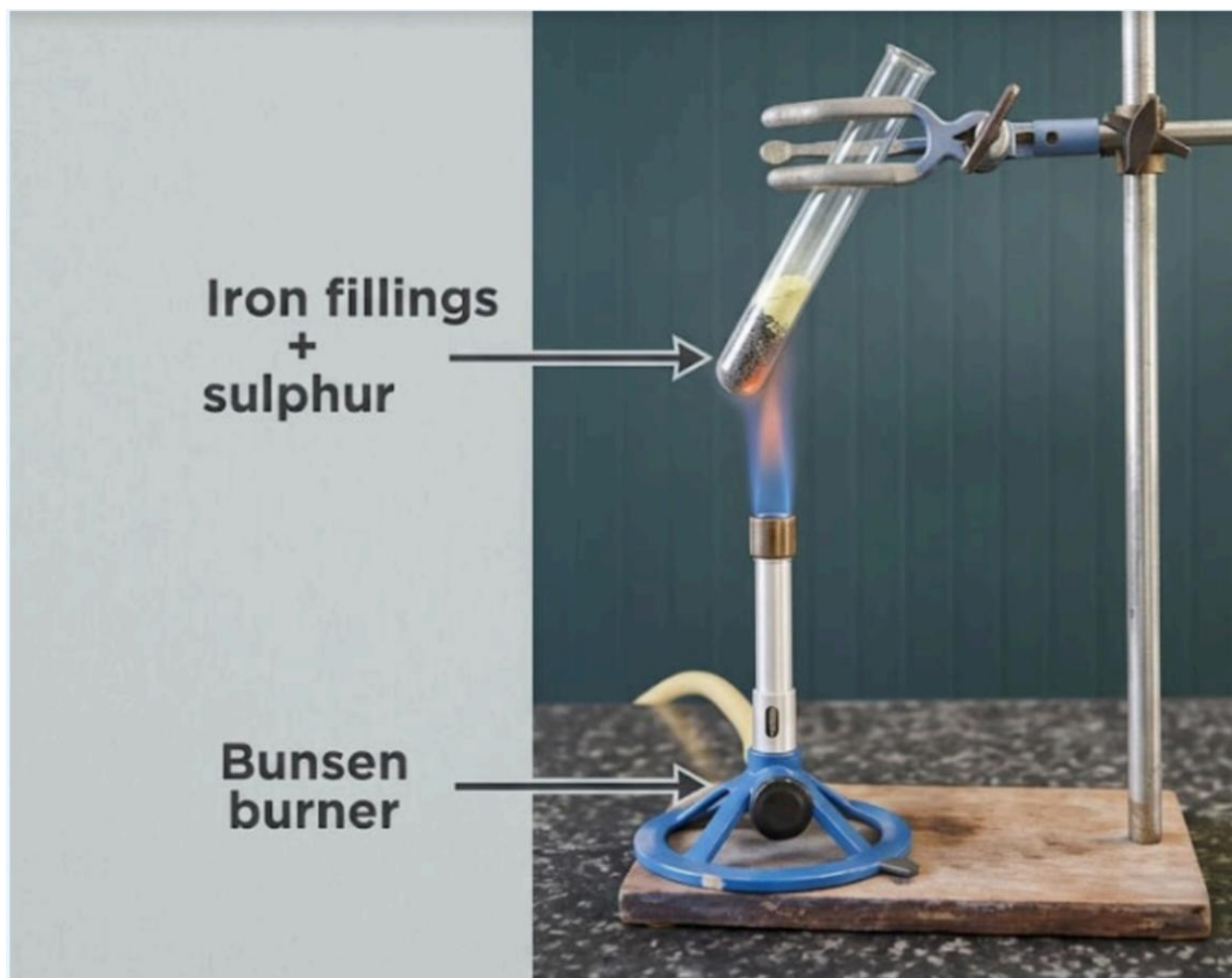
51. Consider the following structure:



Which one of the following is the IUPAC name for the given organic compound?

- 2-Ethyl-6-heptyne
- 6-Ethyl-3-heptyne
- 3-Methyl-5-octyne
- 6-Methyl-3-octyne

52. Which of the following reactions takes place in the above experiment?



- a. Decomposition reaction
- b. Direct combination reaction
- c. Double displacement reaction
- d. Single displacement reaction

53. An equilibrium mixture contains 0.30 mol CO, 0.10 mol H₂, 0.020 mol H₂O and an unknown concentration of CH₄, in 1.0 L container at 1200 K. In this mixture, what is the molar concentration of CH₄ at equilibrium? $\text{CO(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_4\text{(g)} + \text{H}_2\text{O(g)}$, $K_c = 3.92$

- a. 0.059 mol/L
- b. 0.015 mol/L
- c. 0.197 mol/L
- d. 0.001 mol/L

54. Which one of the following is the reason why hydrolysis of sodium carbonate gives a basic solution? This is due to

- a. CO_3^{2-} is hydrolyzed
- b. H^+ is hydrolyzed
- c. OH^- is hydrolyzed
- d. Na^+ is hydrolyzed

55. The thermal decomposition of calcium carbonate during the commercial preparation of lime (CaO) is given below:



What is the atom economy for preparation of lime based on the above reaction? (Atomic masses of Ca=40g/mol, C=12g/mol, O=16g/mol)

- a. 56%
- b. 49.1%
- c. 44%
- d. 100%

56. The rate constant of a reaction increases by a factor of 4 when the temperature is increased from 300K to 320K. What is the approximate activation energy of the reaction? ($R=8.314\text{J}/(\text{mol}\cdot\text{K})$, $\ln(4)\approx 1.386$)

- a. 27.6kJ/mol
- b. 55.3kJ/mol
- c. 110.6kJ/mol
- d. 8.31kJ/mol

57. Calculate the electromotive force (EMF) of the cell $\text{Zn}|\text{Zn}^{2+}(0.01\text{M})||\text{Cu}^{2+}(1.0\text{M})|\text{Cu}$ at 25°C. ($E_{\text{cell}}^\circ=1.10\text{V}$)

- a. 1.10 V
- b. 1.04 V
- c. 1.16 V
- d. 1.22 V

58. What is the pH of a buffer solution prepared by mixing 50.0mL of 0.20M CH_3COOH and 50.0mL of 0.10M NaOH ? (K_a for $\text{CH}_3\text{COOH}=1.8\times 10^{-5}$)

- a. 2.87
- b. 4.74
- c. 5.04
- d. 9.26

59. Calculate the standard enthalpy of formation of liquid benzene C_6H_6 given the following standard enthalpies of combustion: $\text{C}(\text{s})=-394\text{kJ}/\text{mol}$, $\text{H}_2(\text{g})=-286\text{kJ}/\text{mol}$, $\text{C}_6\text{H}_6(\text{l})=-3268\text{kJ}/\text{mol}$

- a. -46 kJ/mol
- b. +46 kJ/mol
- c. +3222 kJ/mol
- d. -3268 kJ/mol

60. Which of the following alkyl halides is most reactive towards an $\text{S}_{\text{N}}1$ substitution reaction mechanism?

- a. 1-Chlorobutane
- b. 2-Chlorobutane
- c. 2-Chloro-2-methylpropane
- d. Chloromethane

Answer key:

1. Answer: C

Explanation: The statement describes **Faraday's second law of electrolysis**, which states that the masses of different substances liberated or deposited by the same quantity of electricity are directly proportional to their equivalent masses.

2. Answer: A

Explanation: The element has electrons filling up to the third principal energy level ($n=3$), so it belongs to **Period 3**. The valence shell configuration is $3s^2 3p^3$, giving it **5 valence electrons**, placing it in **Group VA** (Group 15). The element is Phosphorus.

3. Answer: B

Explanation: Aluminum metal reacts with hydrochloric acid to produce aluminum chloride and hydrogen gas. The balanced chemical equation is: $2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{AlCl}_3\text{(aq)} + 3\text{H}_2\text{(g)}$. Note that the reaction requires 2 Al atoms and 6 HCl molecules to balance the 3 H_2 molecules.

4. Answer: C

Explanation: According to Faraday's second law, the ratio of masses deposited is equal to the ratio of their equivalent weights.

Equivalent weight of Al = $27/3=9$. Equivalent weight of Ag = $108/1=108$.

$$\frac{\text{Mass of Al}}{\text{Eq. wt. of Al}} = \frac{\text{Mass of Ag}}{\text{Eq. wt. of Ag}} \Rightarrow \frac{18}{9} = \frac{\text{Mass of Ag}}{108} \Rightarrow 2 = \frac{\text{Mass of Ag}}{108}$$

Mass of Ag = 216 g.

5. Answer: B

Explanation: Pure nitric acid (HNO_3) is colorless, but over time it undergoes slight decomposition when exposed to light and warmth, producing **nitrogen dioxide (NO_2)**. NO_2 is a reddish-brown gas that dissolves in the acid, giving it a characteristic yellow/brown tint.

6. Answer: D

Explanation: The **molecular formula** tells the exact number of atoms of each element present in a single molecule of a compound. (The empirical formula only gives the simplest whole-number ratio).

7. Answer: B

Explanation: For a weak acid, the concentration of hydrogen ions can be approximated using

$$[\text{H}^+] = \sqrt{K_a \cdot C}$$

$$[\text{H}^+] = \sqrt{(1.4 \times 10^{-5}) \times 0.1} = \sqrt{1.4 \times 10^{-6}} \approx 1.18 \times 10^{-3}\text{M} \approx 1.2 \times 10^{-3}\text{M}$$

8. Answer: C

Explanation: The average atomic mass is calculated as a weighted average of its isotopes:
 $(0.75 \times 35) + (0.25 \times 37) = 26.25 + 9.25 = 35.5$

9. Answer: C

Explanation: In the third test tube, Na_2O_2 (sodium peroxide) is a strong oxidizing agent. In an acidic medium, it oxidizes iodide ions (I^-) from KI to form free iodine (I_2). Iodine reacts with starch solution to produce an intense **blue-black color change**.

(Note: CaO and Al_2O_3 are basic oxides; P_4O_{10} is an acidic oxide but does not act as an oxidizing agent here).

10. Answer: D

Explanation: Acetylene (systematic name: **ethyne**) contains a carbon-carbon triple bond. Hydrocarbons with at least one triple bond belong to the **Alkynes** family.

11. Answer: A

Explanation: Phosphorus pentafluoride (PF_5) is an exception to the octet rule because the

central phosphorus atom shares **10 valence electrons** with the five fluorine atoms, having more than eight valence electrons (an expanded octet).

12. Answer: D

13. Answer: D

Explanation: For the bulb to glow, the aqueous solution must contain free-moving ions (it must be an electrolyte). Sodium hydroxide, copper sulphate, and table salt all dissociate into ions. **Sugar (sucrose)** is a covalent compound that dissolves as intact molecules (a non-electrolyte), so it does not conduct electricity.

14. Answer: C

Explanation: Copper reacts with concentrated sulfuric acid in a redox reaction. The balanced equation is: $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$. Therefore, the coefficient of H_2SO_4 is **2**.

15. Answer: C

Explanation: The energy emitted is calculated using the Rydberg formula: $\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$.

$$E = 2.18 \times 10^{-18} \times \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 2.18 \times 10^{-18} \times \left(\frac{1}{4} - \frac{1}{16} \right).$$

$$E = 2.18 \times 10^{-18} \times \left(\frac{3}{16} \right) \approx 4.09 \times 10^{-19} \text{ J}.$$

16. Answer: D

Explanation] During the electrolytic refining of copper, the impure copper is used as the **anode**. As the copper dissolves, impurities that are less electropositive (like Ag, Au, Pt) fall to the bottom of the cell as sludge, which is known as **anode mud** (labeled 'C').

17. Answer: A

Explanation: The percent ionization of a weak acid is defined as the ratio of the equilibrium concentration of the ionized acid (which is equal to $[\text{H}_3\text{O}^+]$) to the initial concentration of the weak acid, multiplied by 100. Formula: $\frac{\text{Equilibrium concentration of } \text{H}_3\text{O}^+}{\text{Initial concentration of the acid}} \times 100$.

18. Answer: B

Explanation: For the reaction $\text{AB}_2\text{C} + \text{C} \rightarrow \text{AB}_2 + \text{C}_2$ to occur, the incoming free C atom must collide with the C atom bonded inside the AB_2C molecule with the correct spatial orientation to form a new C–C bond. Therefore, the two C atoms must be directly facing each other.

19. Answer: B

Explanation: According to Henry's Law, solubility is directly proportional to partial pressure ($S = k_H \cdot P$).

$$\frac{S_2}{S_1} = \frac{P_2}{P_1} \Rightarrow S_2 = 48.4 \text{ M} \times \left(\frac{0.20 \text{ atm}}{1.0 \text{ atm}} \right) = \mathbf{9.68 \text{ M}}.$$

20. Answer: A

Explanation: Acidic oxides (oxides of non-metals) react with **bases** to form salt and water, with **basic oxides** to form salts, and with **water** to form acids. Because they are already acidic, they do NOT react with acids. Thus, option **A** is incorrect about acidic oxides.

21. Answer: D

Explanation: An element with atomic number 19 is Potassium (K). Its electrons fill the energy levels as **2, 8, 8, 1**. (Notice that it does not fill the 3rd shell completely before starting the 4th shell, as per the specific filling order).

22. Answer: B

Explanation: The invasion of Lake Tana by "Emboch" (Water Hyacinth) causes severe ecological

disruption by covering the surface, blocking sunlight, and depleting oxygen. This is a classic example of **Water pollution**.

23. Answer: B

Explanation: **Democritus**, the ancient Greek philosopher, proposed that matter was made of tiny, indivisible, and discrete particles called "atomos", meaning matter is NOT continuous.

24. Answer: D

Explanation: In normal oxides, oxygen has an oxidation state of -2.

In **peroxides** (like H_2O_2 or Na_2O_2), oxygen has an oxidation state of -1.

25. Answer: B

Explanation: The release of heat during combustion is due to the net change in chemical energy (bond breaking and forming) **that accompanies the reaction process**. It is an inherent property of the thermodynamic change of the reaction, not merely due to light emission or resistance.

26. Answer: A

Explanation: **Teflon** (polytetrafluoroethylene) is an artificial/synthetic polymer created in a laboratory. Enzymes, starch, and cotton are naturally occurring polymers.

27. Answer: D

Explanation: SO_2 has a bent (V-shaped) molecular geometry due to a lone pair on sulfur, causing the dipole moments of the bonds not to cancel out, making it a **polar molecule**. CO_2 , CCl_4 , and CH_4 are all symmetrical and nonpolar.

28. Answer: D

Explanation: According to the Nernst equation (and Le Chatelier's principle applied to electrochemical cells), **increasing the concentration of the reactants** (Ce^{4+}) will shift the equilibrium to the right (product side), which increases the cell potential (voltage).

29. Answer: C

Explanation: In the Contact process for manufacturing sulfuric acid:

- **A** is the furnace where sulfur is burnt in dry air to produce **SO_2** .
- **B** is the catalytic converter where SO_2 is oxidized over V_2O_5 to form **SO_3** .
- **C** is the absorption tower where SO_3 is absorbed in concentrated sulfuric acid (to form oleum, then diluted) to produce **H_2SO_4** .

30. Answer: B

Explanation: **Gun metal** is a specific alloy composed primarily of copper (about 87%), tin (10%), and zinc (3%). It is traditionally used for making gears and gun barrels.

31. Answer: A

Explanation: Given $[\text{OH}^-] = 1.0 \times 10^{-9} \text{M}$, the $\text{pOH} = -\log_{10}(1.0 \times 10^{-9}) = 9$. Since $\text{pH} + \text{pOH} = 14$, the $\text{pH} = 14 - 9 = 5$.

32. Answer: A

Explanation: According to Graham's Law of Diffusion, the rate of diffusion is inversely proportional to the square root of the molar mass (M). Here, $M(\text{H}_2) = 2 \text{ g/mol}$, $M(\text{He}) = 4 \text{ g/mol}$, and $M(\text{O}_2) = 32 \text{ g/mol}$. Since hydrogen is the lightest gas, it diffuses the fastest. Therefore, the **second balloon** (hydrogen) fills first.

33. Answer: B

Explanation: The observation that the total mass of the reactants equals the total mass of the products illustrates the **Law of Conservation of Mass** (proposed by Lavoisier).

34. Answer: A

Explanation: A mixture composed of two or more substances with **at least two visible phases** (physically distinct parts) is defined as a **heterogeneous mixture**.

35. Answer: C

Explanation: Use the combined gas law: $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$.

Given: $P_1 = 60.0 \text{ kPa}$, $V_1 = 50 \text{ cm}^3$, $T_1 = 35 + 273.15 = 308.15 \text{ K}$.

At STP: $P_2 = 101.325 \text{ kPa}$, $T_2 = 273.15 \text{ K}$.

$$V_2 = \frac{60.0 \times 50 \times 273.15}{101.325 \times 308.15} \approx \mathbf{26.24 \text{ cm}^3}.$$

36. Answer: B

Explanation: The structure shown is poly(ethylene terephthalate), commonly known as PET. It is formed by a condensation polymerization reaction between **ethylene glycol** (a diol) and **terephthalic acid** (a dicarboxylic acid).

37. Answer: A

Explanation: 2-hexanol is a 6-carbon straight-chain alcohol with a hydroxyl group (-OH) attached to the second carbon. The correct molecular formula is **CH₃CH₂CH₂CH₂CH(OH)CH₃**.

38. Answer: B

Explanation: **Gold** (Au) is a metallic element whose atoms are held together by metallic bonds, forming a metallic crystal lattice. Diamond is a covalent network crystal; KCl and CaCO₃ are ionic crystals.

39. Answer: D

Explanation: The standard potentials are $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ and $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$. For a standard Zn-Cu cell, the overall cell potential is $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 \text{ V} - (-0.76 \text{ V}) = \mathbf{1.1 \text{ V}}$.

40. Answer: C

Explanation: The factors that affect preferential discharge during electrolysis are: the nature of the ions (their position in the electrochemical series), the concentration of the ions, and the nature of the electrodes. The **physical thickness of the electrodes** does not influence which ion is discharged.

41. Answer: B

Explanation: An element with atomic number 20 is Calcium (Ca), which belongs to Group IIA. It has 2 valence electrons and readily loses them during ionization to achieve a stable octet, forming a **M²⁺** ion.

42. Answer: B

Explanation: In the modern periodic table, Period 1, Period 2, and **Period 3** are known as short periods, each containing 8 or fewer elements. Periods 4 and above are long periods containing transition metals.

43. Answer: A

Explanation: A chemical reaction in which a **single compound breaks down** into two or more simpler components is defined as a **decomposition reaction**.

44. Answer: A

Explanation: According to the law of conservation of mass, the chemical equation must be

balanced with the same number of atoms on both sides. The correctly balanced reaction is: $\text{HCl(aq)} + \text{AgNO}_3\text{(aq)} \rightarrow \text{AgCl(s)} + \text{HNO}_3\text{(l)}$.

45. Answer: B

Explanation: **Gold** is a mineral mined from the earth. Once it is extracted and used, it cannot be regenerated on a human timescale, making it a non-renewable natural resource. Cotton, silk, and wood are renewable biological resources.

46. Answer: D

Explanation: The equilibrium constant (K_c) is a constant for a given reaction at a specific temperature. Adding or removing reactants or products shifts the equilibrium position (Le Chatelier's principle), but it **does NOT change the value of K_c** .

47. Answer: B

Explanation: Phosphorus (P) has 5 valence electrons. In PCl_3 , it forms 3 single bonds with Chlorine atoms and has **1 lone pair of electrons** (AX_3E_1 VSEPR notation). This arrangement causes the bonds to be pushed down, resulting in a **trigonal pyramidal** shape.

48. Answer: B

Explanation: The instantaneous rate of reaction at any point is determined by the steepness (slope) of the tangent to the curve at that point. Since the curve is **steeper at point A** than at point B, the instantaneous rate at point A is greater.

49. Answer: A

Explanation: A neutron is an uncharged (neutral) subatomic particle found in the nucleus of an atom. Therefore, its relative electrical charge is **0**.

50. Answer: A

Explanation: In the diagram, **B** points to the copper electrode. In a Zn-Cu voltaic cell, copper ions (Cu^{2+}) are reduced to copper metal at this electrode, which means it functions as the **cathode**. (Zinc is the anode).

51. Answer: D

Explanation: The longest carbon chain containing the triple bond has 8 carbons (octyne). Numbering from the right gives the triple bond the lowest possible locant (C3). There is a methyl group at C6. Thus, the IUPAC name is **6-methyl-3-octyne**.

52. Answer: B

Explanation: When iron filings and sulfur powder are heated together, they chemically combine to form a single product, iron(II) sulfide. This type of reaction, where two substances combine to form one, is a **direct combination reaction** (or synthesis reaction).

53. Answer: A

Explanation: The equilibrium expression is $K_c = \frac{[\text{CH}_4][\text{H}_2\text{O}]}{[\text{CO}][\text{H}_2]^3}$.

$$3.92 = \frac{[\text{CH}_4](0.020)}{(0.30)(0.10)^3} = \frac{0.020[\text{CH}_4]}{0.0003}$$

$$0.020[\text{CH}_4] = 3.92 \times 0.0003 = 0.001176$$

$$[\text{CH}_4] = \frac{0.001176}{0.020} \approx \mathbf{0.059 \text{ mol/L}}$$

54. Answer: A

Explanation: Sodium carbonate (Na_2CO_3) is the salt of a strong base (NaOH) and a weak acid (H_2CO_3). When dissolved in water, the Na^+ ions are spectator ions, but the **CO_3^{2-} ions hydrolyze** by reacting with water to produce HCO_3^- and OH^- ions, making the solution basic.

55. Answer: A

Explanation: Atom economy = $\frac{\text{Molar mass of desired product}}{\text{Total molar mass of all reactants}} \times 100$.

Molar mass of $\text{CaCO}_3 = 40 + 12 + 48 = 100 \text{ g/mol}$.

Molar mass of desired product (CaO) = $40 + 16 = 56 \text{ g/mol}$.

Atom economy = $\frac{56}{100} \times 100 = 56\%$.

56. Answer: B

Explanation: Use the Arrhenius equation: $\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$.

$\ln(4) \approx 1.386$, $T_1 = 300 \text{ K}$, $T_2 = 320 \text{ K}$, $R = 8.314 \text{ J/(mol} \cdot \text{K)}$.

$1.386 = \frac{E_a}{8.314} \left(\frac{320-300}{300 \times 320}\right) = \frac{E_a}{8.314} \left(\frac{20}{96000}\right)$.

$1.386 = \frac{E_a}{8.314 \times 4800}$.

$E_a = 1.386 \times 8.314 \times 4800 \approx 55,300 \text{ J/mol} = 55.3 \text{ kJ/mol}$.

57. Answer: C

Explanation: Use the Nernst equation: $E = E^\circ - \frac{0.0592}{n} \log\left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}\right)$.

Here, $n = 2$. $E = 1.10 - \frac{0.0592}{2} \log\left(\frac{0.01}{1.0}\right) = 1.10 - 0.0296 \times (-2)$.

$E = 1.10 + 0.0592 \approx 1.16 \text{ V}$.

58. Answer: B

Explanation: Initial moles: $n(\text{CH}_3\text{COOH}) = 0.05 \times 0.20 = 0.010 \text{ mol}$.

$n(\text{NaOH}) = 0.05 \times 0.10 = 0.005 \text{ mol}$.

The NaOH neutralizes the acid, forming 0.005 mol of conjugate base (CH_3COO^-). The remaining acid is $0.010 - 0.005 = 0.005 \text{ mol}$.

Since $[\text{Acid}] = [\text{Base}]$, the Henderson-Hasselbalch equation simplifies to $\text{pH} = \text{pKa}$.

$\text{pKa} = -\log(1.8 \times 10^{-5}) \approx 4.74$.

59. Answer: B

Explanation: Use Hess's Law. The target formation reaction is: $6\text{C(s)} + 3\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{l})$.

Using enthalpies of combustion (ΔH_c):

$\Delta H_f = \sum \Delta H_c(\text{reactants}) - \sum \Delta H_c(\text{products})$.

$\Delta H_f = [6(-394) + 3(-286)] - (-3268)$.

$\Delta H_f = [-2364 - 858] + 3268 = -3222 + 3268 = +46 \text{ kJ/mol}$.

60. Answer: C (3rd option in original list)

Explanation: The rate of an SN_1 reaction depends on the stability of the carbocation intermediate formed in the rate-determining step. A **tertiary carbocation** (formed from 2-chloro-2-methylpropane) is significantly more stable than primary or secondary carbocations due to the inductive and hyperconjugation effects of the three alkyl groups. Therefore, it is the most reactive towards an SN_1 mechanism.

